# PROTON+: Contamination-Robust Prototype Banks for Test-Time Online Out-of-Distribution Detection with Medical Vision-Language Models

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**Abstract**

Medical vision-language models are deployed as zero-shot classifiers and must flag inputs that lie outside the classes they were prompted with. Static detectors such as maximum concept matching (MCM) read only the softmax over the class prompts. An image from a new camera or a new population is classified confidently, so the softmax carries no trace of the shift. PROTON answers this with an online bank of class prototypes filled from confident test predictions, and fuses the distance to the nearest prototype with MCM. We reproduce PROTON from scratch on retinal photographs, chest radiographs and histopathology patches with three medical vision-language models, and find that its confidence gate admits the confidently misclassified shifted inputs, so the prototypes absorb the shift and the detector can fall below MCM. PROTON+ keeps the online bank but anchors every class prototype to the dominant imaging-domain mode of the bank with a mean-shift step, admits only samples that are consistent with their class prototype, and fuses online z-scores instead of raw scores. With the hyper-parameters of PROTON unchanged, PROTON+ improves over MCM and over the best static score on the covariate and far shifts of all three modalities, and its prototype distance alone is the strongest detector on those shifts. On covariate shift PROTON+ reaches 93.9 AUROC on chest radiographs, where MCM is at chance, and gains 16.4 AUROC points over MCM on ultra-widefield fundus photographs.

## 1 Introduction

Medical vision-language models (VLMs) such as `FLAIR`, `UniMed-CLIP` and `QuiltNet` classify an image from text prompts alone, without task-specific training [7, 12, 23]. A screening service prompted with "normal" and "diabetic retinopathy" also receives glaucoma cases, ultra-widefield photographs from a different camera, and images that are not retinas at all. Each of these inputs must be flagged for review instead of being assigned to one of the two classes. Out-of-distribution (OOD) detection assigns every input a score that is higher for inputs the classifier should not decide on, and a detector is judged by how well that score separates in-distribution (ID) inputs from OOD inputs. Benchmarks over many medical datasets report that no single static score works across semantic, covariate and far shifts [5, 9].

The softmax over $C$ class prompts compresses a $d$-dimensional image embedding into $C$ numbers, and a shifted image that the model classifies confidently is indistinguishable from an ID image in that view. Static scores read only this view, whether they take the maximum probability (MCM [19]), the maximum logit or the energy [6, 17], add patch-level matches (GL-MCM [21]), or average over augmented prompts (TAG [18]). Hierarchical prompts enrich the text side and improve detection on medical VLMs, at the price of expert-written prompt trees [9]. Few-shot methods learn prompts or alignments from labelled ID

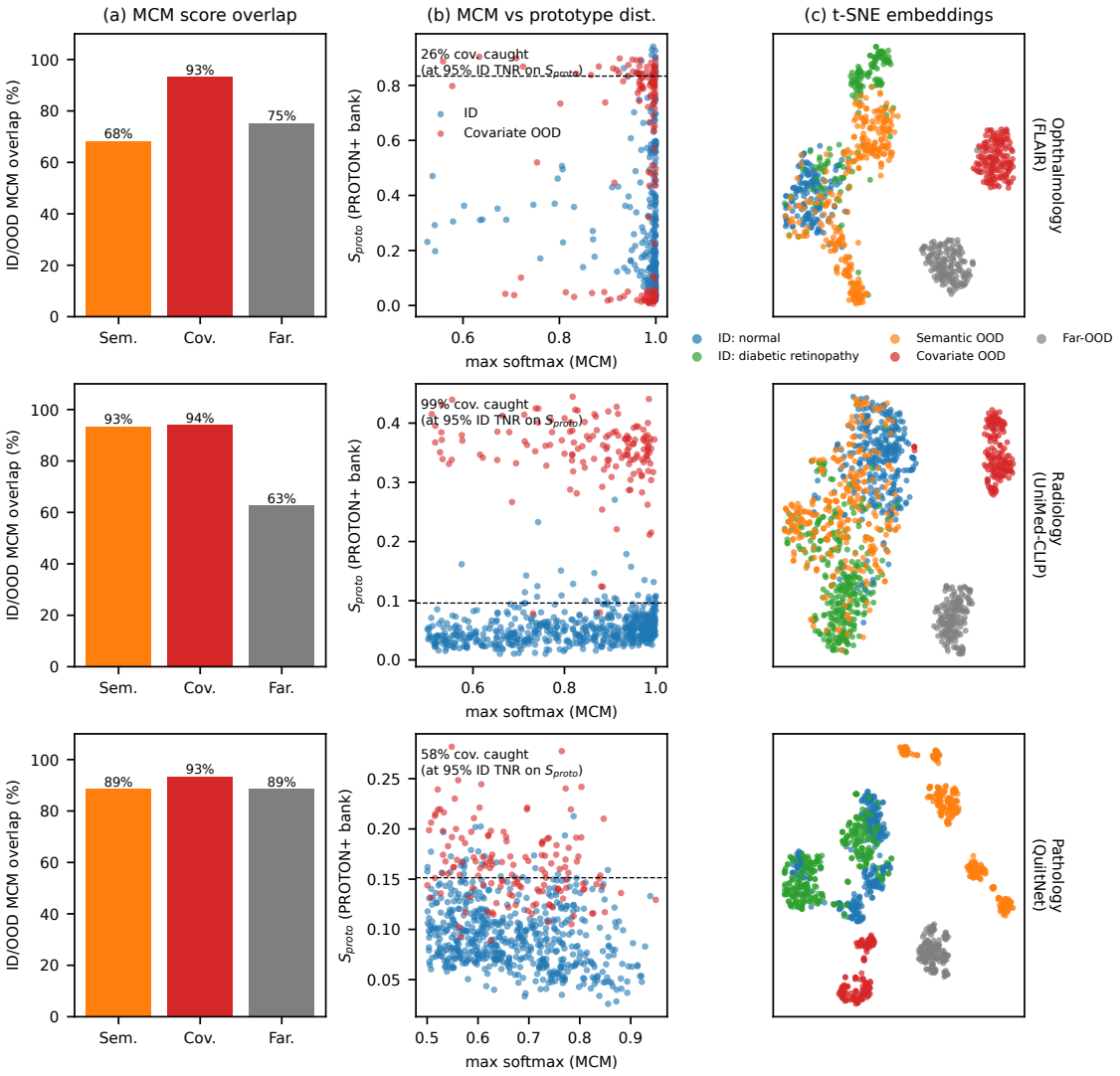



Figure 1: **Shifted inputs are separable in embedding space but not in softmax space.** (a) share of OOD MCM scores inside the 5th to 95th percentile range of ID scores, (b) MCM confidence against the online prototype distance on the covariate stream with the fraction of covariate samples above the 95th ID percentile, (c) t-SNE of the embeddings. Rows: `FLAIR`, `UniMed-CLIP`, `QuiltNet`.

images [14, 20, 27, 31]. Test-time methods adapt a VLM from the unlabelled stream itself, by learning textual OOD concepts (TTL [30]), negative semantics (TINS [29]) or prompts from a memory of confident samples (STAR [4]), and they update text embeddings or prompts. PROTON [2] is the closest work. It keeps the VLM frozen, maintains an online bank of class prototypes from confident predictions, and fuses the distance to the nearest prototype with MCM. Its evidence is a single ophthalmology benchmark, and whether the mechanism survives an independent reproduction is open.

Shifted inputs occupy regions of the embedding space that ID inputs do not, while their MCM scores overlap with ID scores, as Figure 1 shows for three modalities. The prototype bank therefore reads the right signal, but the confidence gate is the wrong filter. In our reproduction 97% of the ultra-widefield images pass the gate, 148 of 150 are assigned to the diabetic-retinopathy class, and that class bank ends with 63% shifted members, so the prototype moves towards the shift it should detect. A stricter gate does not help, because the

shifted images are as confident as the ID images. PROTON+ replaces the filter with two label-free statistics of the bank itself. A mean-shift step over all bank members finds the dominant imaging-domain mode, and each class prototype is the mean of its members weighted by their proximity to that mode, so a confidently classified but shifted cluster receives almost no weight. A sample enters the bank only if its distance to the prototype of its predicted class is typical of the samples admitted before it. The two scores are fused as online z-scores, which removes the scale mismatch between a probability and a cosine distance.

We evaluate on three benchmarks with semantic, covariate and far-OOD splits: retinal photographs with `FLAIR`, chest radiographs with `UniMed-CLIP`, and histopathology patches with `QuiltNet`, against seven zero-shot scores and two detectors that use labelled ID features. PROTON+ improves over MCM on covariate and far shifts in every modality, and on semantic shift in radiology and pathology. On the ultra-widefield covariate shift, where PROTON loses 22.1 AUROC points against MCM, PROTON+ gains 16.4 points. The prototype distance alone, computed from the PROTON+ bank, reaches 90 to 99 AUROC on covariate and far shifts, and fusing it with a weak base score is what caps the fused detector. The MCM baseline itself is 27 points above the value the original paper reports on the ultra-widefield split, so the gain over MCM, not the absolute PROTON score, is what fails to transfer. Both PROTON and PROTON+ assume that ID inputs form the majority of the stream, and their scores invert when shifted inputs make up half of it or more.

Our contributions are:

1. We reproduce PROTON from scratch on three modalities and release the library, manifests and cached features. The reported gain over MCM does not transfer, because the MCM baseline is 27 to 42 points higher than reported, and on the ultra-widefield covariate shift PROTON scores 22.1 AUROC points below it.

2. We trace the failure to bank contamination by confidently misclassified shifted inputs, and show that the variance-driven fusion weight stays between 0.42 and 0.65 on every stream, so it is a constant in practice.

3. We introduce PROTON+, which anchors prototypes to the dominant imaging-domain mode, gates bank admission by prototype consistency, and fuses z-scores, at 0.1 to 0.3 ms per sample on one CPU core and 0.4 MB of state.

4. We quantify the ceiling of the prototype signal with an oracle bank that never admits a shifted sample, and the sensitivity of every detector to the gate, the bandwidth and the prompt wording, which moves MCM by up to 28 AUROC points between single-prompt wordings.

## 2 Related Work

**Zero-shot OOD scores for vision-language models.** MCM scores an image by the maximum softmax probability over the class prompts, scaled by the temperature of the contrastive model [19]. Maximum-logit and energy scores read the unnormalised similarities instead of the probabilities [6, 17]. GL-MCM adds the best match between any image patch and any class prompt [21], and TAG averages the maximum probability over randomly shuffled prompt wordings [18]. All of these are stateless, so two images with the same probabilities receive the same score regardless of where they lie in the embedding space [28].

**OOD detection for medical vision-language models.** Ju et al. build a cross-modality benchmark with semantic, covariate and far shifts for `FLAIR`, `UniMed-CLIP` and `QuiltNet`, show that no static score wins on all shifts, and propose hierarchical prompts that describe anatomy, modality and lesion morphology [9]. OpenMIBOOD reaches the same conclusion over fourteen datasets [5]. Hierarchical

vision-language learning and global-local alignment improve medical OOD detection with labelled ID images [14, 27]. Our work uses the ophthalmology benchmark of Ju et al. and builds the radiology and pathology benchmarks from public datasets with the same three shift types.

**Test-time adaptation and memory banks.** Prompt learning adapts a VLM from a few labelled images [31], and LoCoOp regularises the prompts with local background features for OOD detection [20]. Test-time methods drop the labels and learn from the unlabelled stream: TTL updates textual prompts with pseudo-labelled test samples and a bank of textual features [30], TINS learns negative text embeddings that are kept apart from ID prototypes [29], and STAR stores confident samples in a memory bank to build class prototypes for prompt adaptation [4]. These methods update text embeddings or prompts and target the accuracy of the classifier on the shifted inputs. PROTON keeps the model frozen and uses an online bank of image prototypes only to score inputs [2]. Our work keeps the bank and the fusion of PROTON and changes how prototypes are estimated and which samples may enter the bank.

**Feature-bank detectors with labels.** Nearest-neighbour and Mahalanobis distances to labelled ID features are strong detectors when such features are available [15, 24]. We report them as an upper bound that none of the zero-shot methods may use.

# 3 Method

PROTON+ scores every image of an unlabelled stream with a frozen VLM, an online bank of class prototypes, and a fusion of a softmax score with a prototype distance. We first state the setting and PROTON, then show why its bank gets contaminated, and then describe the three changes of PROTON+, which are domain-anchored prototypes, consistency admission and scale-free fusion.

## 3.1 Setting and PROTON

A frozen VLM maps an image $x_t$ to an $\ell_2$-normalised embedding $e_t \in \mathbb{R}^d$ and the $C$ class names to text embeddings $T = [t_1, \dots, t_C]$, computed once. The zero-shot classifier is $p_t = \mathrm{softmax}(e_t^\top T/\tau)$ with the temperature $\tau$ learned by the VLM, and MCM scores $S_{\mathrm{MCM}}(x_t) = -\max_c p_{t,c}$, so that a higher score means more OOD. Images arrive one at a time, no labels and no training data are available, and the detector may use only the images it has already seen.

PROTON [2] keeps one first-in-first-out queue $\mathcal{D}_c$ of at most $M$ embeddings per class. After an image is scored, it enters the queue of its predicted class $\hat{c} = \arg\max_c p_{t,c}$ if $\max_c p_{t,c} \geq \gamma$, and the oldest member leaves when the queue is full. The prototype of class $c$ is the normalised mean of its queue,

$$\mu_c = \bar{e}_c / \|\bar{e}_c\|_2, \qquad \bar{e}_c = \frac{1}{|\mathcal{D}_c|} \sum_{e \in \mathcal{D}_c} e, \tag{1}$$

and the prototype distance is one minus the cosine similarity to the nearest prototype,

$$S_{\mathrm{proto}}(x_t) = 1 - \max_c e_t^\top \mu_c. \tag{2}$$

Once every queue holds at least $K_{\min}$ members, the score is the convex combination

$$S_t = \alpha_t \, S_{\mathrm{MCM}}(x_t) + (1 - \alpha_t) \, S_{\mathrm{proto}}(x_t), \qquad \alpha_t = \alpha_{\max} - \sigma\big(100 \, (\sigma_t^2 - \sigma_0^2)\big) \, (\alpha_{\max} - \alpha_{\min}), \tag{3}$$

where $\sigma_t^2$ is the running variance of $S_{\mathrm{MCM}}$ over the stream, maintained with Welford's algorithm [26], $\sigma_0^2$ is a reference variance, $\sigma(\cdot)$ is the logistic function, and $\alpha_t$ moves from $\alpha_{\max}$ towards $\alpha_{\min}$ when the variance of $S_{\mathrm{MCM}}$ rises above the reference. Before calibration the score is $S_{\mathrm{MCM}}$ alone.

### 3.2 Why the bank gets contaminated

The gate $\max_c p_{t,c} \geq \gamma$ is meant to keep OOD embeddings out of the bank. It can only exclude inputs with a low maximum probability, which are exactly the inputs that MCM already detects. A covariate-shifted input is one that the model classifies confidently for the wrong reason, so it passes the gate by definition and lands in the bank of the class it was assigned to. When such inputs are a sizeable share of the stream, the prototype of that class becomes the mean of two clusters, the ID cluster and the shifted cluster, and lies between them. Every member of either cluster is then at a similar distance from the prototype, and Equation (2) no longer separates them. The fusion in Equation (3) adds a second problem, because $S_{\mathrm{MCM}}$ lies in $[-1, -1/C]$ while $S_{\mathrm{proto}}$ lies in $[0, 2]$, so the weight $\alpha_t$ does not control the relative influence of the two signals.

### 3.3 PROTON+

**Domain-anchored prototypes.** All ID classes of a benchmark share one imaging domain, so the union of all queues has one dominant mode whenever ID members outnumber the shifted members of the union. This is a heuristic that holds for the streams with an ID fraction of at least $0.7$ in our data and fails below that, as the experiments show. A mean-shift step [1] over the union finds that mode. Starting from the normalised mean $\mu$ of all members $\mathcal{D} = \bigcup_c \mathcal{D}_c$, we iterate

$$\mu \leftarrow \frac{\sum_{e \in \mathcal{D}} w(e, \mu)\, e}{\left\| \sum_{e \in \mathcal{D}} w(e, \mu)\, e \right\|_2}, \qquad w(e, \mu) = \exp\big((e^\top \mu - 1)/h\big), \tag{4}$$

three times, where $h$ is a bandwidth on the cosine scale and $w$ down-weights members that are far from the current mode. The result $\mu_{\mathrm{dom}}$ is the domain anchor. Each class prototype is then the domain-weighted mean of its own queue,

$$\mu_c = \frac{\sum_{e \in \mathcal{D}_c} w(e, \mu_{\mathrm{dom}})\, e}{\left\| \sum_{e \in \mathcal{D}_c} w(e, \mu_{\mathrm{dom}})\, e \right\|_2}, \tag{5}$$

so a member whose cosine similarity to the anchor is lower by $h$ than that of a typical ID member contributes $e^{-1}$ as much, and a member in a shifted cluster at cosine distance several times $h$ contributes almost nothing. The prototype distance is still Equation (2).

**Consistency admission.** After calibration a sample enters the queue of its predicted class $\hat{c}$ only if its confidence passes the gate and its distance $d_t = 1 - e_t^\top \mu_{\hat{c}}$ is typical of the samples admitted before it,

$$d_t \leq m_t + \kappa\, s_t, \tag{6}$$

where $m_t$ and $s_t$ are the running mean and standard deviation of the distances of samples admitted after calibration, also maintained with Welford's algorithm, and $\kappa$ sets the tolerance. The rule becomes active once $K_{\min}$ such distances have been recorded, so the first $K_{\min}$ confident samples after calibration are admitted unconditionally and seed $m_t$ and $s_t$. The rule costs one dot product and three scalar updates, and it rejects the members that Equation (5) would have down-weighted, so the queue fills with ID samples and the FIFO eviction removes early contamination over time.

**Scale-free fusion.** We keep Equation (3) with the same weight $\alpha_t$ but replace each score by its online z-score, $(S_{\mathrm{MCM}} - \bar{m}_t)/\hat{s}_t$ and $(S_{\mathrm{proto}} - \bar{p}_t)/\hat{q}_t$. The statistics of $S_{\mathrm{MCM}}$ run over every sample seen so far, and those of $S_{\mathrm{proto}}$ over the samples seen since calibration, because the prototype distance does not exist before. Before calibration the z-scored MCM is emitted, so the score keeps one scale across the calibration boundary. A variant takes the maximum of the two z-scores instead of their weighted sum, which flags an input when either signal is anomalous and has no fusion weight.

**Cost.** Per sample PROTON+ computes $C$ dot products for the prototype distance and one for the admission rule. Each admission triggers the anchor update of Equation (4), which is two matrix-vector products

per iteration over the $CM \times d$ bank matrix plus one per class for Equation (5), so the cost of an admission grows with $CMd$ and the number of iterations. The state is the bank of $CM$ single-precision embeddings and a handful of scalars, 0.4 MB for $C = 2$, $M = 100$ and $d = 512$. No gradient, augmentation or text re-encoding is involved, and any score from Section 2 can replace MCM as the base signal. Two variants are evaluated alongside: the maximum of the two z-scores in place of the weighted sum, and a fusion weight driven by the rejection rate of the admission rule instead of the variance of MCM, with the same bounds $[\alpha_{\min}, \alpha_{\max}]$.

# 4 Experiments

**Benchmarks.** We use three benchmarks, each with two ID classes, three OOD splits and one VLM, and we follow the construction of Ju et al. where it exists [9]. *Ophthalmology* uses FIVES [8] with normal and diabetic retinopathy as ID classes (150 images each), glaucoma and age-related macular degeneration from the same dataset as the semantic shift (150 each), ultra-widefield fundus photographs from DeepDRiD [16] as the covariate shift (150), and natural images from the Imagenette subset of ImageNet [3] as the far shift (150). *Radiology* uses the RSNA pneumonia challenge [22] with normal and lung-opacity radiographs as ID classes (300 each), NIH ChestX-ray14 radiographs whose findings lie outside the ID classes as the semantic shift (300), paediatric radiographs [11] of both classes as the covariate shift (150), and ImageNet as the far shift (150). *Pathology* uses PatchCamelyon [25] with benign and malignant lymph-node patches as ID classes (300 each), five colorectal tissue types from NCT-CRC-HE [10] with no counterpart in the ID classes as the semantic shift (300), patches of both classes from the held-out hospital of Camelyon17-WILDS [13] as the covariate shift (150), and ImageNet as the far shift (150). Every image list is sampled once with a fixed seed and released with the code. A disjoint labelled ID split, 50 images per class in ophthalmology and 300 per class in the other two benchmarks, serves only the two detectors that need labelled features.

**Models and prompts.** The frozen VLMs are `FLAIR` (ResNet-50 and Bio_ClinicalBERT, $d = 512$) [23], `UniMed-CLIP` (ViT-B/16 and BiomedBERT) [12] and `QuiltNet` (ViT-B/32) [7], with the temperature each model learned. Prompts follow the public benchmark code: `FLAIR` uses its built-in caption "A fundus photograph of {class}" and `UniMed-CLIP` uses "x-ray of {class}". No benchmark code exists for PatchCamelyon, so for `QuiltNet` we chose "normal" and "cancer" among twelve wordings by zero-shot accuracy on the disjoint labelled split, which never touches the evaluated images. `FLAIR` averages text-encoder states over all tokens of a padded batch, so the class prompts are encoded as one batch, exactly as its forward pass does. The zero-shot accuracies on the ID test images are 84.0%, 69.2% and 72.7%, and the appendix reports every detector under every prompt set.

**Streams and protocol.** A stream contains all ID test images and all images of one OOD split in a random order, so the ID fraction is 0.50, 0.67 and 0.67 for the three splits in ophthalmology and 0.67, 0.80 and 0.80 in radiology and pathology. Every online detector is run over five random orders, all numbers for online detectors are the mean $\pm$ standard deviation over these five orders, and paired comparisons count the orders on which a detector beats MCM on the same stream. Static scores do not depend on the order. We report AUROC, the area under the precision-recall curve with OOD as the positive class (AUPR), and the false-positive rate on ID images at 95% OOD recall (FPR@95), all in percent. PROTON runs with the values of its paper, $\gamma = 0.7$, $M = 100$, $K_{\min} = 5$, $\alpha_{\min} = 0.3$, $\alpha_{\max} = 0.7$ and $\sigma_0^2 = 0.02$. PROTON+ adds $h = 0.1$, $\kappa = 2$ and three mean-shift iterations, fixed before the experiments from the typical within-class cosine spread and the two-sigma rule. The sweeps in the appendix show that these defaults are not the best values in every modality. Features are extracted once per benchmark on one GPU in under a minute, and every detector then runs on a CPU.

**Baselines.** The zero-shot baselines are Max-Logits, Energy, MCM, GL-MCM, TAG with the class prompt and ten random word shuffles of it, which reduces to MCM for the single-word pathology prompts,

Table 1: **PROTON does not reproduce on the ophthalmology benchmark, PROTON+ recovers covariate and far shifts.** Zero-shot OOD detection with `FLAIR`. AUROC and AUPR higher is better, FPR@95 lower is better. Online detectors report mean ± SD over five stream orders. † hierarchical prompts of Ju et al. Best per column in bold, second best underlined.

| | Semantic | | | Covariate | | | Far-OOD | | |
|---|---|---|---|---|---|---|---|---|---|
| Method | AUROC↑ | AUPR↑ | FPR@95↓ | AUROC↑ | AUPR↑ | FPR@95↓ | AUROC↑ | AUPR↑ | FPR@95↓ |
| Max-Logits | 54.0 | 54.1 | <u>79.3</u> | 27.9 | 23.3 | 88.0 | 83.9 | 76.4 | 63.0 |
| Energy | 51.8 | 51.5 | 79.7 | 27.2 | 23.2 | 88.0 | 83.9 | <u>76.5</u> | 62.7 |
| GL-MCM | **75.8** | **75.6** | 86.0 | 68.9 | 45.3 | 69.3 | 78.5 | 62.4 | 75.0 |
| MCM | <u>75.8</u> | <u>75.6</u> | 86.0 | 68.9 | 45.2 | 69.7 | 78.5 | 62.4 | 75.0 |
| TAG-MCM | 68.4 | 70.5 | 88.7 | 30.1 | 23.8 | 93.7 | 72.2 | 47.2 | 60.3 |
| MCM (L=1)† | 66.5 | 65.0 | 84.7 | <u>84.2</u> | **66.3** | <u>47.3</u> | 51.8 | 38.4 | 98.3 |
| MCM (L=5)† | 58.3 | 56.5 | 87.0 | 66.5 | 42.5 | 65.7 | 54.4 | 33.5 | 74.3 |
| PROTON [2] | 68.0 | 72.5 | 92.7 | 46.7 | 29.4 | 83.6 | <u>86.8</u> | 76.2 | <u>55.0</u> |
| PROTON+ (ours) | 67.3 | 69.0 | **75.8** | **85.3** | <u>63.9</u> | **31.5** | **93.4** | **81.8** | **15.4** |
| Δ PROTON+ vs. MCM | ▼8.4 | ▼6.5 | ▲10.2 | ▲16.4 | ▲18.6 | ▲38.1 | ▲14.9 | ▲19.4 | ▲59.6 |

and the hierarchical prompts of Ju et al. at one and five descriptions per class. Entropy equals MCM for two classes and is omitted. kNN and Mahalanobis distances to the labelled ID split are reported as privileged upper bounds. The following subsections ask whether PROTON reproduces, why it fails, what PROTON+ adds, which of its parts matter, how sensitive it is, and how it behaves when shifted inputs dominate the stream.

### 4.1 Does PROTON reproduce?

Table 1 lists every zero-shot score on the ophthalmology benchmark, with the AUROC, AUPR and FPR@95 of each shift, and Table 2 gives the three modalities. A row above the MCM row on AUROC beats the static baseline the online detectors build on. **PROTON improves on far shifts in every modality and on the covariate shift in radiology and pathology, but it loses** 22.1 **AUROC points to MCM on the ultra-widefield covariate shift and** 7.7 **points on the semantic shift of the same benchmark.** Over the calibrated part of all 45 streams the running variance of MCM stays between 0.0001 and 0.029, so with $\sigma_0^2 = 0.02$ the fusion weight ranges only from 0.42 to 0.65 and the adaptive fusion behaves as a constant. The confidence gate costs rather than helps in radiology and pathology, where the bank without a gate scores 6.5 and 2.2 points higher on covariate shift, as Table 3 shows for ophthalmology and the appendix for the other two modalities. The original paper reports 62.7, 66.3 and 84.5 AUROC for PROTON and 42.4 for MCM on the covariate split [2]. With the benchmark prompt of that paper our PROTON reaches 66.6 on the same split, within 0.3 points of the reported value, but MCM reaches 84.2 with that prompt and 68.9 with the default caption, so the reported gain over MCM is what does not transfer. The standard deviation over stream orders is 0.8 to 3.6 AUROC points and the per-order range reaches 11 points on the ophthalmology far shift.

### 4.2 Why does it fail? Bank contamination

Figure 1 plots, per modality, the overlap of MCM scores between ID and OOD, the MCM confidence against the prototype distance, and a t-SNE of the embeddings. Covariate and far samples form clusters away from the ID classes in every modality, while their MCM scores overlap with ID scores. The gate admits what it should exclude. On the ultra-widefield stream, averaged over the five orders, 97% of the shifted images

Table 2: **PROTON+ improves over MCM on covariate and far shifts in every modality.** AUROC, AUPR and FPR@95 per shift for the three benchmarks, mean ± SD over five stream orders for the online detectors.

| | | Semantic | | | Covariate | | | Far-OOD | | |
|---|---|---|---|---|---|---|---|---|---|---|
| Domain (VLM) | Method | AUROC | AUPR | FPR@95 | AUROC | AUPR | FPR@95 | AUROC | AUPR | FPR@95 |
| Ophthalmology (FLAIR) | MCM | 75.8 | 75.6 | 86.0 | 68.9 | 45.2 | 69.7 | 78.5 | 62.4 | 75.0 |
| | PROTON | 68.0 ± 0.9 | 72.5 ± 0.7 | 92.7 ± 2.3 | 46.7 ± 1.8 | 29.4 ± 0.8 | 83.6 ± 6.1 | 86.8 ± 3.6 | 76.2 ± 2.6 | 55.0 ± 26.0 |
| | PROTON+ | 67.3 ± 0.7 | 69.0 ± 0.3 | 75.8 ± 1.7 | 85.3 ± 7.8 | 63.9 ± 6.8 | 31.5 ± 26.5 | 93.4 ± 0.5 | 81.8 ± 1.2 | 15.4 ± 1.8 |
| Radiology (UniMed-CLIP) | MCM | 54.2 | 35.2 | 89.7 | 46.6 | 18.1 | 87.3 | 27.6 | 14.0 | 99.7 |
| | PROTON | 55.3 ± 0.9 | 38.3 ± 0.5 | 95.2 ± 1.2 | 61.2 ± 1.7 | 31.0 ± 1.4 | 67.6 ± 3.7 | 93.9 ± 2.3 | 92.5 ± 2.1 | 33.5 ± 33.5 |
| | PROTON+ | 61.4 ± 0.7 | 42.4 ± 0.5 | 88.1 ± 1.4 | 93.9 ± 0.6 | 82.1 ± 3.4 | 26.7 ± 5.4 | 91.5 ± 1.4 | 74.6 ± 4.4 | 35.7 ± 31.5 |
| Pathology (QuiltNet) | MCM | 48.8 | 33.1 | 97.2 | 48.0 | 19.4 | 92.0 | 45.1 | 18.7 | 95.0 |
| | PROTON | 52.9 ± 0.8 | 36.5 ± 0.8 | 94.0 ± 1.3 | 55.6 ± 1.4 | 28.7 ± 1.4 | 90.3 ± 3.6 | 58.2 ± 2.9 | 33.1 ± 3.6 | 89.2 ± 5.3 |
| | PROTON+ | 72.5 ± 2.0 | 57.8 ± 2.9 | 73.8 ± 2.7 | 70.1 ± 1.6 | 44.1 ± 1.7 | 76.5 ± 1.9 | 81.1 ± 1.1 | 61.9 ± 1.4 | 64.3 ± 4.5 |

pass $\gamma = 0.7$ with a mean confidence of $0.955$ against $0.970$ for ID images, 148 of 150 are assigned to the diabetic-retinopathy class, and that queue ends with $63\%$ shifted members. On the paediatric radiograph stream the normal queue ends with $59\%$ shifted members, and on the pathology semantic stream the benign queue with $54\%$. **With an oracle admission rule that never admits a shifted sample, PROTON separates the ultra-widefield images from ID at** $93.1$ **AUROC instead of** $46.7$ **and the paediatric radiographs at** $86.5$ **instead of** $61.2$**, so in these two modalities the signal is there and the filter is the fault.** In pathology the same oracle lifts PROTON only to $61.4$ on covariate shift, so there the prototype estimate and the fusion scale are the larger faults.

## 4.3 What does PROTON+ add?

Table 2 compares MCM, PROTON and PROTON+ on the three modalities. **PROTON+ beats MCM on the covariate and far shifts of every modality and on all five orders of every such stream, by** $16.4$ **to** $47.3$ **AUROC points on covariate shift and by** $14.9$ **to** $63.9$ **points on far shift, and it beats PROTON on seven of the nine shift-modality pairs, trailing by** $0.7$ **points on the ophthalmology semantic shift and by** $2.4$ **points on the radiology far shift.** MCM is at chance on four of these six shifts, so against the best static score of each split the margins are $1.1$, $15.5$ and $4.0$ points on covariate shift and $9.4$, $7.0$ and $14.0$ points on far shift. On semantic shift PROTON+ gains $7.2$ points over MCM in radiology and $23.7$ points in pathology, and loses $8.4$ points in ophthalmology, where the glaucoma and macular-degeneration images lie inside the ID imaging domain and MCM already separates them at $75.8$. Against the best static score it trails by $8.4$ and $8.8$ points on the ophthalmology and radiology semantic shifts and leads by $12.7$ points in pathology. FPR@95 on the ultra-widefield shift drops from $69.7$ for MCM and $83.6$ for PROTON to $31.5$, and on the paediatric shift from $87.3$ and $67.6$ to $26.7$. The AUROC standard deviation over stream orders is at most $2.0$ points except on the ultra-widefield shift ($7.8$), where the 11 to 21 samples before calibration decide how much of the shifted cluster is admitted before the anchor forms, and the FPR@95 standard deviation is large on two shifts ($26.5$ ultra-widefield, $31.5$ radiology far). The pathology gains hold under the other prompt wordings too: with the literal class names PROTON+ reaches $75.8$, $72.9$ and $92.5$ against $64.3$, $53.4$ and $72.5$ for MCM, and with the histopathology template $82.4$, $85.6$ and $92.1$ against $59.8$, $66.1$ and $67.1$.

## 4.4 Which parts matter?

Table 3 adds the three changes one at a time on the ophthalmology benchmark, and Table 4 lists the prototype score on its own and the oracle bank for every modality. Z-scoring alone lifts covariate shift by $4$ to $17$ points, and far shift by $2$ and $11$ points in ophthalmology and pathology at a cost of $3$ points in radiology,

Table 3: **Domain anchoring carries the covariate gain.** Ablation on the ophthalmology benchmark (AUROC). The top block adds the components of PROTON as in its paper, the bottom block adds the changes of PROTON+ one at a time.

| Variant | Sem | Cov | Far |
|---|---|---|---|
| MCM baseline | 75.8 | 68.9 | 78.5 |
| + Proto. ($\alpha$=.5) | 65.5 | 46.6 | 86.9 |
| + Adapt. fusion | 65.3 | 46.9 | 85.3 |
| + Conf. gate (Full) | 68.0 | 46.7 | 86.8 |
| PROTON + z-scored fusion | 69.4 | 50.7 | 89.0 |
| + consistency admission | 70.6 | 40.1 | 92.6 |
| + domain-anchored prototypes (PROTON+) | 67.3 | 85.3 | 93.4 |

Table 4: **The prototype signal, not the fusion, is the detector.** AUROC (mean $\pm$ SD over five stream orders) of the prototype distance alone from the PROTON+ bank, and of PROTON+ with an oracle admission rule that never admits a shifted sample.

| Modality | Variant | Sem | Cov | Far |
|---|---|---|---|---|
| Ophth. | PROTON+ | $67.3 \pm 0.7$ | $85.3 \pm 7.8$ | $93.4 \pm 0.5$ |
| | $S_{\text{proto}}$ only (PROTON+ bank) | $57.0 \pm 1.0$ | $90.4 \pm 12.6$ | $97.9 \pm 0.8$ |
| | PROTON+, oracle admission | $73.6 \pm 2.9$ | $90.4 \pm 1.1$ | $93.5 \pm 0.6$ |
| Radio. | PROTON+ | $61.4 \pm 0.7$ | $93.9 \pm 0.6$ | $91.5 \pm 1.4$ |
| | $S_{\text{proto}}$ only (PROTON+ bank) | $62.6 \pm 0.5$ | $98.8 \pm 0.5$ | $96.2 \pm 2.2$ |
| | PROTON+, oracle admission | $64.9 \pm 0.9$ | $92.4 \pm 1.5$ | $91.3 \pm 1.6$ |
| Patho. | PROTON+ | $72.5 \pm 2.0$ | $70.1 \pm 1.6$ | $81.1 \pm 1.1$ |
| | $S_{\text{proto}}$ only (PROTON+ bank) | $91.0 \pm 1.9$ | $90.1 \pm 2.1$ | $98.2 \pm 0.9$ |
| | PROTON+, oracle admission | $76.2 \pm 1.7$ | $72.0 \pm 1.3$ | $79.6 \pm 1.5$ |

because the two raw scores have different variances and the weight $\alpha_t$ cannot compensate for that. Consistency admission alone cannot repair a bank whose seed is already mixed, and on the ultra-widefield shift it lowers the AUROC to $40.1$. **Domain-anchored prototypes carry the covariate gain: they raise the ultra-widefield shift from** $40.1$ **to** $85.3$ **and the paediatric shift from** $78.4$ **to** $93.9$**.** The prototype distance computed from the PROTON+ bank is the strongest single signal on covariate and far shifts, with $90.4$, $98.8$ and $90.1$ AUROC on the three covariate shifts and $96$ to $98$ on the far shifts, and it beats the fused score in pathology on every shift. Fusion with MCM pays only where MCM carries information the embedding lacks, which is the ophthalmology semantic shift ($57.0$ alone against $67.3$ fused). An oracle admission rule that never admits a shifted sample is a reference rather than a bound, because it calibrates later and PROTON+ exceeds it on three pairs by up to $1.5$ points. In ophthalmology it reaches $73.6$, $90.4$ and $93.5$, so PROTON+ is within $6.3$ points of it on semantic and $5.1$ points on covariate shift, and within $3.7$ points on every other pair. The maximum of the two z-scores replaces the fusion weight and gains $5$ to $14$ points in pathology, where MCM is near chance, and changes the other six pairs by $-5$ to $+1$ points. A fusion weight driven by the rejection rate of the admission rule beats the variance-driven weight on seven of nine pairs, by up to $12$ points on the pathology far shift, and loses $2.9$ points on the ophthalmology semantic shift.

## 4.5 How sensitive is it?

Table 5 sweeps the gate $\gamma$ and the bank size $M$ for PROTON, and the appendix sweeps $\gamma$, $\kappa$, $h$ and the prompt wording for PROTON+. **PROTON+ moves by at most** $2.7$ **points for** $\kappa \in [1, 4]$ **and by at most** $3.4$ **points for** $\gamma \in [0.5, 0.7]$ **in every modality, and the defaults are not the best values, because** $\kappa = 1$ **or** $h = 0.05$ **improves on them on six of nine pairs each, by up to** $2.4$ **and** $4.9$ **points.** A stricter gate starves the bank where the ID images are less confident. With mean ID confidences of $0.82$ in radiology and $0.67$ in pathology, $\gamma = 0.8$ costs $5$ to $13$ points in pathology, and $\gamma = 0.9$ drops the pathology shifts to chance and the radiology covariate and far shifts by $16$ and $20$ points. A bandwidth of $h \leq 0.1$ works in every modality, $h = 0.2$ loses $13$ points on the ultra-widefield shift and $12$ points on the pathology semantic shift, and $h = 0.5$ no longer separates the ultra-widefield cluster from the ID mode. Prompt wording moves the static baseline more than any detector choice: a trailing period in the `FLAIR` prompt moves the MCM covariate AUROC from $68.9$ to $84.2$ and the far AUROC from $78.5$ to $51.8$, while PROTON+ stays between $85.3$ and $94.0$ on covariate and between $88.1$ and $97.0$ on far shift across the three single-prompt wordings. On covariate and far shifts any base score benefits from the bank, and with Max-Logits or Energy, which score $27$ to $28$ on the ultra-widefield shift, PROTON+ reaches $87$ on covariate and $99$ on far shift.

Table 5: **PROTON is stable in $M$ but its covariate AUROC rises with a stricter gate, because fewer shifted images are admitted.** Ultra-widefield covariate shift, PROTON with the defaults of its paper otherwise.

| $\gamma$ | AUC | APR | F95 | $M$ | AUC | APR | F95 |
|---|---|---|---|---|---|---|---|
| 0.5 | 46.9 | 29.4 | 82.5 | 10 | 45.3 | 29.3 | 92.0 |
| 0.6 | 46.7 | 29.4 | 83.8 | 25 | 44.6 | 28.6 | 89.9 |
| 0.7 | 46.7 | 29.4 | 83.6 | 50 | 45.4 | 28.9 | 87.2 |
| 0.8 | 47.8 | 30.1 | 81.3 | 100 | 46.7 | 29.4 | 83.6 |
| 0.9 | 49.4 | 30.7 | 79.9 | 200 | 47.9 | 29.9 | 79.5 |

Table 6: **Online prototypes need an ID majority.** AUROC against the ID fraction $\rho$ of the ophthalmology stream (Sem / Cov / Far).

| | MCM | | | PROTON | | | PROTON+ | | |
|---|---|---|---|---|---|---|---|---|---|
| $\rho$ | Sem | Cov | Far | Sem | Cov | Far | Sem | Cov | Far |
| 90% | 76.2 | 67.7 | 75.5 | 73.1 | 85.5 | 90.4 | 70.6 | 88.5 | 90.9 |
| 70% | 74.8 | 69.0 | 79.0 | 71.2 | 47.7 | 87.4 | 68.7 | 74.7 | 92.9 |
| 50% | 75.8 | 69.1 | 78.4 | 68.2 | 34.0 | 55.3 | 67.4 | 22.3 | 36.7 |
| 30% | 76.2 | 68.3 | 78.0 | 60.3 | 28.0 | 26.4 | 64.5 | 28.8 | 27.8 |
| 10% | 75.9 | 69.2 | 78.1 | 50.0 | 41.3 | 16.7 | 56.8 | 41.4 | 20.5 |

### 4.6 When shifted inputs dominate the stream

Table 6 varies the ID fraction $\rho$ of the stream by sub-sampling the smaller pool, so its rows are not the streams of Table 2. **Both detectors assume an ID majority: at $\rho = 0.9$ and $0.7$ PROTON+ beats MCM on covariate and far shifts in every modality, and once the shifted cluster dominates the bank the prototype score inverts.** The inversion starts at $\rho = 0.5$ on the ophthalmology covariate and far shifts and on the pathology covariate shift, and at $\rho = 0.3$ elsewhere, where the ultra-widefield AUROC of PROTON+ is $22.3$ at $\rho = 0.5$ and the far-OOD AUROC is $20.5$ at $\rho = 0.1$, against $69$ and $78$ for MCM, which is order-free. PROTON degrades in the same way, with $34.0$ and $16.7$ at the same points, so the claim of its paper that it never falls below MCM does not hold for streams with an ID minority.

### 4.7 Cost

One PROTON step takes $0.02$ ms and one PROTON+ step $0.10$ to $0.28$ ms on a single CPU core in NumPy, the difference being the anchor update after every admission, and the state is $0.41$ MB for $C = 2$, $M = 100$ and $d = 512$. Every queue holds $K_{\min} = 5$ members, and the detector is calibrated, after 14 to 20 samples on average in ophthalmology, 30 to 74 in radiology and 33 to 37 in pathology, depending on the split.

## 5 Discussion

**The prototype signal is the detector.** On every covariate and far shift the prototype distance from the PROTON+ bank beats the fused score, and on seven of nine pairs it beats every static baseline. The fusion of the original design protects the semantic shift of a modality whose embedding keeps the shifted classes inside the ID domain, as in ophthalmology, and costs up to 20 points elsewhere. The variance of MCM does not indicate a shift. The rejection rate of the admission rule is a better statistic on seven of nine pairs, and

neither statistic recovers the ophthalmology semantic loss, so choosing the weight from the stream without labels remains open.

**Towards streams with an ID minority.** Both detectors invert once the shifted cluster dominates the bank, because the dominant mode is then the shift. The text prompts are the only label-free anchor of the ID domain, and in our data the shifted images are as close to the prompts as the ID images, so a text-anchored mode would not help. A deployment that cannot guarantee an ID majority needs a small labelled seed set, which turns the bank into the nearest-neighbour detector of the appendix.

**Towards shift-aware benchmarks.** Every covariate and far split in the three benchmarks, and in the benchmark of Ju et al., comes from a different dataset than the ID split, and the privileged detectors separate those splits at 99 to 100 AUROC. The benchmarks therefore measure how well a detector recovers dataset identity from the embedding without labels, which is the quantity test-time prototypes target, and they do not isolate a shift type from an acquisition difference. Splits drawn from the same acquisition pipeline, as our radiology semantic split from NIH ChestX-ray14, are the step towards that.

**Towards test-time learners as baselines.** TTL, TINS and STAR update text embeddings or prompts from the stream and were not run here, because they target the accuracy of the adapted classifier rather than a frozen detector. The released streams and features make that comparison a direct next step.

**Towards robust prompts.** A trailing period moves the MCM covariate AUROC of `FLAIR` by 15 points, and the text embedding of a prompt depends on the other prompts in its batch. Any comparison of zero-shot detectors on medical VLMs should report the prompt wording and the encoding protocol, and PROTON+ reduces but does not remove this dependence.

# 6 Conclusion

We reproduce PROTON, an online prototype-bank detector for medical vision-language models, on three modalities and find that its confidence gate admits the confidently misclassified inputs it is meant to exclude, so the prototypes absorb the shift. PROTON+ anchors the prototypes to the dominant imaging-domain mode of the bank, admits only samples consistent with their class prototype, and fuses z-scores. With the same hyper-parameters it improves over MCM on covariate and far shifts in every modality, by up to 47.3 and 63.9 AUROC points, by up to 15.5 and 14.0 points over the best static score of a split, and it stays within 6.3 AUROC points of an oracle admission rule on every shift. The library, the image manifests, the cached features and a browser demo with the real `FLAIR` weights are released so that every number in this paper can be recomputed.

**AI assistance.** An AI coding assistant (Claude) implemented the library, ran the experiments and drafted the text under the authors' direction, and the authors reviewed the code, the protocol and every number.

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# A Benchmark construction

**Ophthalmology.** FIVES holds 800 fundus photographs of 2048 by 2048 pixels, 200 per class, split by the dataset into training and test folders that we merge. Per ID class we draw 150 images for the test stream and keep the remaining 50 as the labelled split, and per semantic class we draw 150. The 150 ultra-widefield images are drawn from the 256 photographs of the three DeepDRiD ultra-widefield folders. ImageNet images are 150 validation images of the ten-class Imagenette subset at full resolution.

**Radiology.** The RSNA images are the DICOM radiographs of the challenge training set, converted to 8-bit PNG with per-image min-max scaling. We draw 600 per class and use the first 300 as the test stream and the rest as the labelled split. The semantic split is drawn from the public 5,606-image sample of NIH ChestX-ray14 after removing every image whose findings include No Finding, Pneumonia, Infiltration, Consolidation, Effusion or Edema, so that cardiomegaly, pneumothorax, masses, nodules, atelectasis and fibrosis remain. RSNA is itself a subset of NIH ChestX-ray14, so every NIH candidate whose 64 by 64 grayscale correlation with an RSNA image of the ID or labelled split exceeds $0.95$ is removed before sampling, which drops 22 of the 1,834 candidates, while unrelated RSNA radiographs reach at most $0.95$ among themselves. The paediatric split draws 75 normal and 75 pneumonia images from the test folder of Kermany et al. A second semantic split of 300 COVID-19 radiographs from the COVID-19 Radiography Database is released with the manifests but not used in the tables, because COVID-19 pneumonia overlaps the pneumonia class.

**Pathology.** PatchCamelyon patches are 96 by 96 pixels. We draw 600 per label from the test split, 300 for the stream and 300 for the labelled split. The semantic split draws 60 patches from each of adipose, debris, mucus, smooth muscle and cancer-associated stroma in NCT-CRC-HE-7K. Normal mucosa, lymphocytes, tumour epithelium and background are excluded because they resemble the ID classes or carry no tissue. The covariate split draws 75 patches per label from the out-of-distribution test hospital of Camelyon17-WILDS.

# B Prompt sets

Tables 7 to 9 report every prompt set with its zero-shot ID accuracy and the AUROC of MCM, PROTON and PROTON+. “vanilla” is the bare class name, “l1” the one-description template of Ju et al. (for `FLAIR` the built-in caption with a trailing period, for `UniMed-CLIP` the benchmark prompt, for `QuiltNet` a histopathology template), and “l5” the five-description hierarchy of Ju et al., which we wrote for radiology and pathology in the same style.

Table 7: Prompt sensitivity, ophthalmology (AUROC Sem / Cov / Far).

| Prompt set | ID acc. | MCM (Sem/Cov/Far) | PROTON | PROTON+ |
|---|---|---|---|---|
| flair_default | 84.0 | 75.8 / 68.9 / 78.5 | 68.0 / 46.7 / 86.8 | 67.3 / 85.3 / 93.4 |
| vanilla | 86.0 | 65.6 / 64.9 / 83.5 | 58.5 / 46.7 / 94.4 | 58.2 / 87.9 / 97.0 |
| l1 | 83.7 | 66.5 / 84.2 / 51.8 | 65.0 / 66.6 / 78.2 | 64.6 / 94.0 / 88.1 |
| l5 | 57.3 | 58.3 / 66.5 / 54.4 | 41.2 / 24.0 / 25.3 | 56.2 / 50.8 / 58.8 |

Table 8: Prompt sensitivity, radiology.

| Prompt set | ID acc. | MCM (Sem/Cov/Far) | PROTON | PROTON+ |
|---|---|---|---|---|
| vanilla | 73.2 | 41.8 / 57.5 / 55.5 | 44.9 / 78.9 / 94.7 | 53.2 / 95.5 / 96.2 |
| l1 | 69.2 | 54.2 / 46.6 / 27.6 | 55.3 / 61.2 / 93.9 | 61.4 / 93.9 / 91.5 |
| l5 | 55.3 | 54.2 / 78.4 / 84.6 | 50.3 / 79.1 / 88.3 | 49.9 / 88.5 / 96.8 |

Table 9: Prompt sensitivity, pathology.

| Prompt set | ID acc. | MCM (Sem/Cov/Far) | PROTON | PROTON+ |
|---|---|---|---|---|
| id_acc | 72.7 | 48.8 / 48.0 / 45.1 | 52.9 / 55.6 / 58.2 | 72.5 / 70.1 / 81.1 |
| vanilla | 64.5 | 64.3 / 53.4 / 72.5 | 66.7 / 58.9 / 82.2 | 75.8 / 72.9 / 92.5 |
| l1 | 59.3 | 59.8 / 66.1 / 67.1 | 64.9 / 73.0 / 75.2 | 82.4 / 85.6 / 92.1 |
| l5 | 65.5 | 15.3 / 64.6 / 61.1 | 32.6 / 61.5 / 65.7 | 15.4 / 73.4 / 81.5 |

# C Full tables for radiology and pathology

# D Broader baselines and base scores

Table 15 lists TAG, the five-description hierarchy, the two privileged detectors and PROTON+, and Table 16 instantiates PROTON and PROTON+ on six base scores. The privileged detectors use the labelled split and separate every OOD split at 99 to 100 AUROC on covariate and far shifts. The hierarchy of Ju et al. with PROTON on top inverts in ophthalmology, because the five-description prompts classify the ID images at $57\%$ accuracy and the bank fills with misrouted samples, and PROTON+ collapses under the same prompts for the same reason.

# E Sensitivity of PROTON+

Table 17 sweeps the mean-shift bandwidth $h$, and Figure 2 the gate and bank size of PROTON on all three modalities. The contamination statistics of Section 4.2 are means over the five stream orders, and the trajectories of Figure 3 come from the first order. The reference variance $\sigma_0^2$ of the fusion weight is not inert. Sweeping it from $0.005$ to $0.08$ moves the mean AUROC of PROTON over the three shifts by 2 points in ophthalmology, rising with $\sigma_0^2$, and by $14$ points in radiology and $8$ points in pathology, falling with $\sigma_0^2$. A three-way triage of the shift type from the pair of scores reaches a macro-F1 of $0.60$ in ophthalmology, $0.85$ in radiology and $0.46$ in pathology with a logistic regression under five-fold cross-validation.

# F Text encoding in `FLAIR`

The text tower of `FLAIR` averages the hidden states of three BERT layers over all token positions of the padded batch, padding included. The embedding of "A fundus photograph of normal" therefore changes with the length of the longest prompt in its batch. The forward pass of the released model encodes all class prompts in one batch, and the benchmark notebooks of Ju et al. call that forward pass, so our main protocol encodes the class prompts jointly. Encoding each prompt alone, as the model's own helper for text

Table 10: Zero-shot OOD detection with `UniMed-CLIP` on the radiology benchmark.

| | Semantic | | | Covariate | | | Far-OOD | | |
|---|---|---|---|---|---|---|---|---|---|
| Method | AUROC↑ | AUPR↑ | FPR@95↓ | AUROC↑ | AUPR↑ | FPR@95↓ | AUROC↑ | AUPR↑ | FPR@95↓ |
| Max-Logits | 69.4 | 56.1 | 82.3 | 0.4 | 10.8 | 100.0 | 75.7 | 64.6 | 90.2 |
| Energy | **70.2** | **57.4** | 84.8 | 0.4 | 10.8 | 100.0 | 77.0 | 67.4 | 90.5 |
| GL-MCM | 54.3 | 35.7 | 89.7 | 46.7 | 18.1 | 87.3 | 29.2 | 14.2 | 98.0 |
| MCM | 54.2 | 35.2 | 89.7 | 46.6 | 18.1 | 87.3 | 27.6 | 14.0 | 99.7 |
| TAG-MCM | 57.4 | 36.8 | 87.8 | 48.1 | 19.3 | 85.5 | 41.3 | 16.0 | 87.5 |
| MCM (L=1)† | 54.2 | 35.2 | 89.7 | 46.6 | 18.1 | 87.3 | 27.6 | 14.0 | 99.7 |
| MCM (L=5)† | 54.2 | 32.0 | **77.5** | 78.4 | 38.2 | 46.3 | 84.6 | 52.3 | 39.5 |
| PROTON [2] | 55.3 | 38.3 | 95.2 | 61.2 | 31.0 | 67.6 | **93.9** | **92.5** | **33.5** |
| PROTON+ (ours) | 61.4 | 42.4 | 88.1 | **93.9** | **82.1** | **26.7** | 91.5 | 74.6 | 35.7 |
| Δ PROTON+ vs. MCM | ▲7.2 | ▲7.1 | ▲1.5 | ▲47.3 | ▲64.1 | ▲60.7 | ▲63.9 | ▲60.6 | ▲64.0 |

Table 11: Zero-shot OOD detection with `QuiltNet` on the pathology benchmark.

| | Semantic | | | Covariate | | | Far-OOD | | |
|---|---|---|---|---|---|---|---|---|---|
| Method | AUROC↑ | AUPR↑ | FPR@95↓ | AUROC↑ | AUPR↑ | FPR@95↓ | AUROC↑ | AUPR↑ | FPR@95↓ |
| Max-Logits | 7.1 | 19.6 | 99.2 | 24.5 | 12.8 | 98.2 | 17.7 | 11.9 | 99.0 |
| Energy | 6.7 | 19.5 | 99.2 | 23.9 | 12.7 | 97.3 | 17.5 | 11.9 | 98.8 |
| GL-MCM | 46.5 | 31.2 | 97.7 | 46.0 | 18.1 | 96.8 | 44.2 | 18.4 | 96.8 |
| MCM | 48.8 | 33.1 | 97.2 | 48.0 | 19.4 | 92.0 | 45.1 | 18.7 | 95.0 |
| TAG-MCM | 48.7 | 33.1 | 97.2 | 48.0 | 19.4 | 92.0 | 45.1 | 18.7 | 95.0 |
| MCM (L=1)† | 59.8 | 42.4 | 90.3 | 66.1 | 29.8 | 85.2 | 67.1 | 31.5 | 77.2 |
| MCM (L=5)† | 15.3 | 20.4 | 99.8 | 64.6 | 30.7 | 77.3 | 61.1 | 26.3 | 87.3 |
| PROTON [2] | 52.9 | 36.5 | 94.0 | 55.6 | 28.7 | 90.3 | 58.2 | 33.1 | 89.2 |
| PROTON+ (ours) | **72.5** | **57.8** | **73.8** | **70.1** | **44.1** | **76.5** | **81.1** | **61.9** | **64.3** |
| Δ PROTON+ vs. MCM | ▲23.7 | ▲24.7 | ▲23.4 | ▲22.2 | ▲24.7 | ▲15.5 | ▲36.0 | ▲43.2 | ▲30.7 |

embeddings does, moves the MCM covariate AUROC by 7 points with the default caption and by 22 points with the bare class names.

Table 12: Ablation, radiology (AUROC).

| Variant | Sem | Cov | Far |
|---|---|---|---|
| MCM baseline | 54.2 | 46.6 | 27.6 |
| + Proto. ($\alpha$=.5) | 55.2 | 66.0 | 96.0 |
| + Adapt. fusion | 55.3 | 67.7 | 97.2 |
| + Conf. gate (Full) | 55.3 | 61.2 | 93.9 |
| PROTON + z-scored fusion | 61.6 | 78.1 | 90.5 |
| + consistency admission | 61.9 | 78.4 | 92.8 |
| + domain-anchored prototypes (PROTON+) | 61.4 | 93.9 | 91.5 |

Table 13: Ablation, pathology (AUROC).

| Variant | Sem | Cov | Far |
|---|---|---|---|
| MCM baseline | 48.8 | 48.0 | 45.1 |
| + Proto. ($\alpha$=.5) | 56.7 | 61.1 | 68.5 |
| + Adapt. fusion | 54.4 | 57.8 | 62.5 |
| + Conf. gate (Full) | 52.9 | 55.6 | 58.2 |
| PROTON + z-scored fusion | 59.6 | 64.5 | 69.2 |
| + consistency admission | 53.8 | 65.9 | 70.9 |
| + domain-anchored prototypes (PROTON+) | 72.5 | 70.1 | 81.1 |

Table 14: ID fraction sweep, radiology (top) and pathology (bottom), AUROC Sem / Cov / Far.

| | MCM | | | PROTON | | | PROTON+ | | |
|---|---|---|---|---|---|---|---|---|---|
| $\rho$ | Sem | Cov | Far | Sem | Cov | Far | Sem | Cov | Far |
| 90% | 53.7 | 46.8 | 27.5 | 57.8 | 71.0 | 94.1 | 64.8 | 96.4 | 94.1 |
| 70% | 54.4 | 47.2 | 27.9 | 54.9 | 54.3 | 81.8 | 62.0 | 89.5 | 81.0 |
| 50% | 54.5 | 47.9 | 28.5 | 54.5 | 49.9 | 58.8 | 58.8 | 69.0 | 73.0 |
| 30% | 54.5 | 46.7 | 27.7 | 54.0 | 38.8 | 37.1 | 55.6 | 12.2 | 14.2 |
| 10% | 53.5 | 47.5 | 27.5 | 53.8 | 23.7 | 21.4 | 52.5 | 12.7 | 13.1 |

| | MCM | | | PROTON | | | PROTON+ | | |
|---|---|---|---|---|---|---|---|---|---|
| $\rho$ | Sem | Cov | Far | Sem | Cov | Far | Sem | Cov | Far |
| 90% | 49.0 | 45.9 | 46.0 | 65.9 | 56.9 | 67.0 | 85.3 | 74.5 | 87.8 |
| 70% | 48.9 | 48.5 | 45.6 | 54.7 | 50.7 | 52.7 | 72.5 | 62.5 | 73.8 |
| 50% | 48.5 | 48.1 | 45.4 | 47.9 | 46.6 | 49.7 | 50.1 | 42.0 | 55.2 |
| 30% | 49.4 | 48.7 | 45.7 | 45.2 | 40.3 | 43.8 | 32.1 | 27.9 | 39.1 |
| 10% | 45.9 | 44.7 | 41.9 | 33.5 | 31.6 | 39.7 | 15.9 | 24.7 | 28.3 |

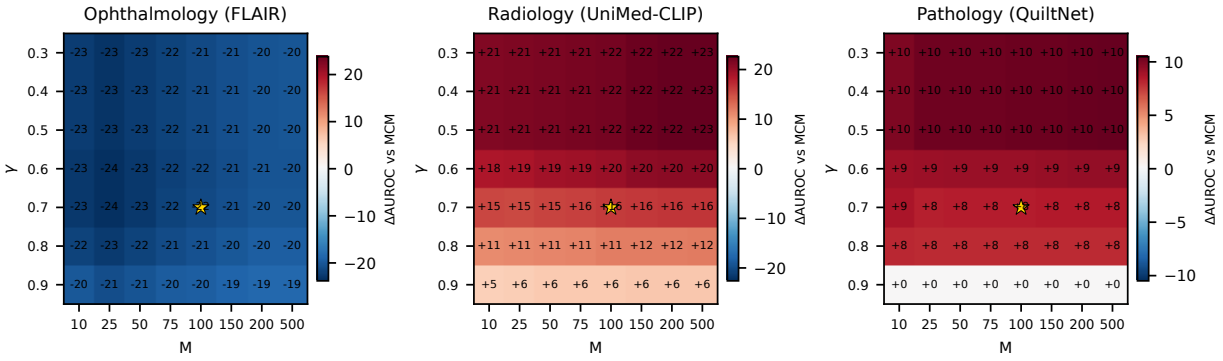



Figure 2: **A stricter gate helps PROTON only where ID images are more confident than the shifted ones.** Change in covariate AUROC of PROTON over MCM against $\gamma$ and $M$, with a star at the defaults.

Table 15: Broader baselines (AUROC Sem / Cov / Far). ‡ uses labelled ID features.

| Ophthalmology | | | |
|---|---|---|---|
| Method | Sem | Cov | Far |
| TAG-MCM | 68.4 | 30.1 | 72.2 |
| MCM (L=5)† | 58.3 | 66.5 | 54.4 |
| kNN-ID‡ | 76.8 | 100.0 | 99.8 |
| Mahalanobis-ID‡ | 82.4 | 100.0 | 100.0 |
| PROTON | 68.0 | 46.7 | 86.8 |
| MCM (L=5) + PROTON | 41.2 | 24.0 | 25.3 |
| PROTON+ (ours) | 67.3 | 85.3 | 93.4 |
| **Radiology** | | | |
| Method | Sem | Cov | Far |
| TAG-MCM | 57.4 | 48.1 | 41.3 |
| MCM (L=5)† | 54.2 | 78.4 | 84.6 |
| kNN-ID‡ | 67.8 | 99.9 | 100.0 |
| Mahalanobis-ID‡ | 58.9 | 100.0 | 100.0 |
| PROTON | 55.3 | 61.2 | 93.9 |
| MCM (L=5) + PROTON | 50.3 | 79.1 | 88.3 |
| PROTON+ (ours) | 61.4 | 93.9 | 91.5 |
| **Pathology** | | | |
| Method | Sem | Cov | Far |
| TAG-MCM | 48.7 | 48.0 | 45.1 |
| MCM (L=5)† | 15.3 | 64.6 | 61.1 |
| kNN-ID‡ | 100.0 | 99.4 | 100.0 |
| Mahalanobis-ID‡ | 100.0 | 99.8 | 100.0 |
| PROTON | 52.9 | 55.6 | 58.2 |
| MCM (L=5) + PROTON | 32.6 | 61.5 | 65.7 |
| PROTON+ (ours) | 72.5 | 70.1 | 81.1 |

Table 16: Five base scores alone, with PROTON and with PROTON+ (AUROC Sem / Cov / Far), ophthalmology, radiology and pathology from top to bottom.

| | Base | | | +PROTON | | | +PROTON+ | | |
|---|---|---|---|---|---|---|---|---|---|
| Base score | Sem | Cov | Far | Sem | Cov | Far | Sem | Cov | Far |
| Max-Logits | 54.0 | 27.9 | 83.9 | 57.3 | 23.6 | 87.0 | 57.1 | 87.1 | 99.1 |
| Energy | 51.8 | 27.2 | 83.9 | 55.3 | 22.6 | 87.0 | 56.1 | 87.2 | 99.1 |
| GL-MCM | 75.8 | 68.9 | 78.5 | 67.6 | 45.7 | 85.0 | 67.3 | 85.3 | 93.4 |
| MCM | 75.8 | 68.9 | 78.5 | 68.0 | 46.7 | 86.8 | 67.3 | 85.3 | 93.4 |
| TAG-MCM | 68.4 | 30.1 | 72.2 | 67.9 | 22.6 | 89.4 | 65.2 | 58.7 | 91.5 |

| | Base | | | +PROTON | | | +PROTON+ | | |
|---|---|---|---|---|---|---|---|---|---|
| Base score | Sem | Cov | Far | Sem | Cov | Far | Sem | Cov | Far |
| Max-Logits | 69.4 | 0.4 | 75.7 | 67.1 | 4.5 | 80.6 | 69.1 | 92.9 | 98.8 |
| Energy | 70.2 | 0.4 | 77.0 | 67.8 | 4.5 | 81.8 | 69.2 | 92.9 | 98.9 |
| GL-MCM | 54.3 | 46.7 | 29.2 | 55.1 | 60.6 | 93.5 | 61.4 | 93.9 | 91.6 |
| MCM | 54.2 | 46.6 | 27.6 | 55.3 | 61.2 | 93.9 | 61.4 | 93.9 | 91.5 |
| TAG-MCM | 57.4 | 48.1 | 41.3 | 58.2 | 60.6 | 94.2 | 63.6 | 86.8 | 83.9 |

| | Base | | | +PROTON | | | +PROTON+ | | |
|---|---|---|---|---|---|---|---|---|---|
| Base score | Sem | Cov | Far | Sem | Cov | Far | Sem | Cov | Far |
| Max-Logits | 7.1 | 24.5 | 17.7 | 11.3 | 29.3 | 24.6 | 75.7 | 81.5 | 93.7 |
| Energy | 6.7 | 23.9 | 17.5 | 10.9 | 28.6 | 24.5 | 75.6 | 81.9 | 93.6 |
| GL-MCM | 46.5 | 46.0 | 44.2 | 51.2 | 54.2 | 58.0 | 71.7 | 70.1 | 80.6 |
| MCM | 48.8 | 48.0 | 45.1 | 52.9 | 55.6 | 58.2 | 72.5 | 70.1 | 81.1 |
| TAG-MCM | 48.7 | 48.0 | 45.1 | 52.9 | 55.6 | 58.2 | 72.5 | 70.1 | 81.1 |

Table 17: Bandwidth $h$ of the domain anchor (AUROC).

| | Ophthalmology | | | Radiology | | | Pathology | | |
|---|---|---|---|---|---|---|---|---|---|
| $h$ | Sem | Cov | Far | Sem | Cov | Far | Sem | Cov | Far |
| 0.02 | 66.9 | 82.7 | 89.2 | 58.5 | 94.4 | 91.8 | 76.7 | 69.1 | 78.8 |
| 0.05 | 68.1 | 86.9 | 92.5 | 60.7 | 94.1 | 91.6 | 77.4 | 70.8 | 80.2 |
| 0.1 | 67.3 | 85.3 | 93.4 | 61.4 | 93.9 | 91.5 | 72.5 | 70.1 | 81.1 |
| 0.2 | 66.9 | 72.6 | 93.8 | 61.6 | 94.2 | 91.6 | 60.8 | 68.7 | 81.0 |
| 0.5 | 68.1 | 40.3 | 93.7 | 61.8 | 86.4 | 92.0 | 55.8 | 67.3 | 73.4 |

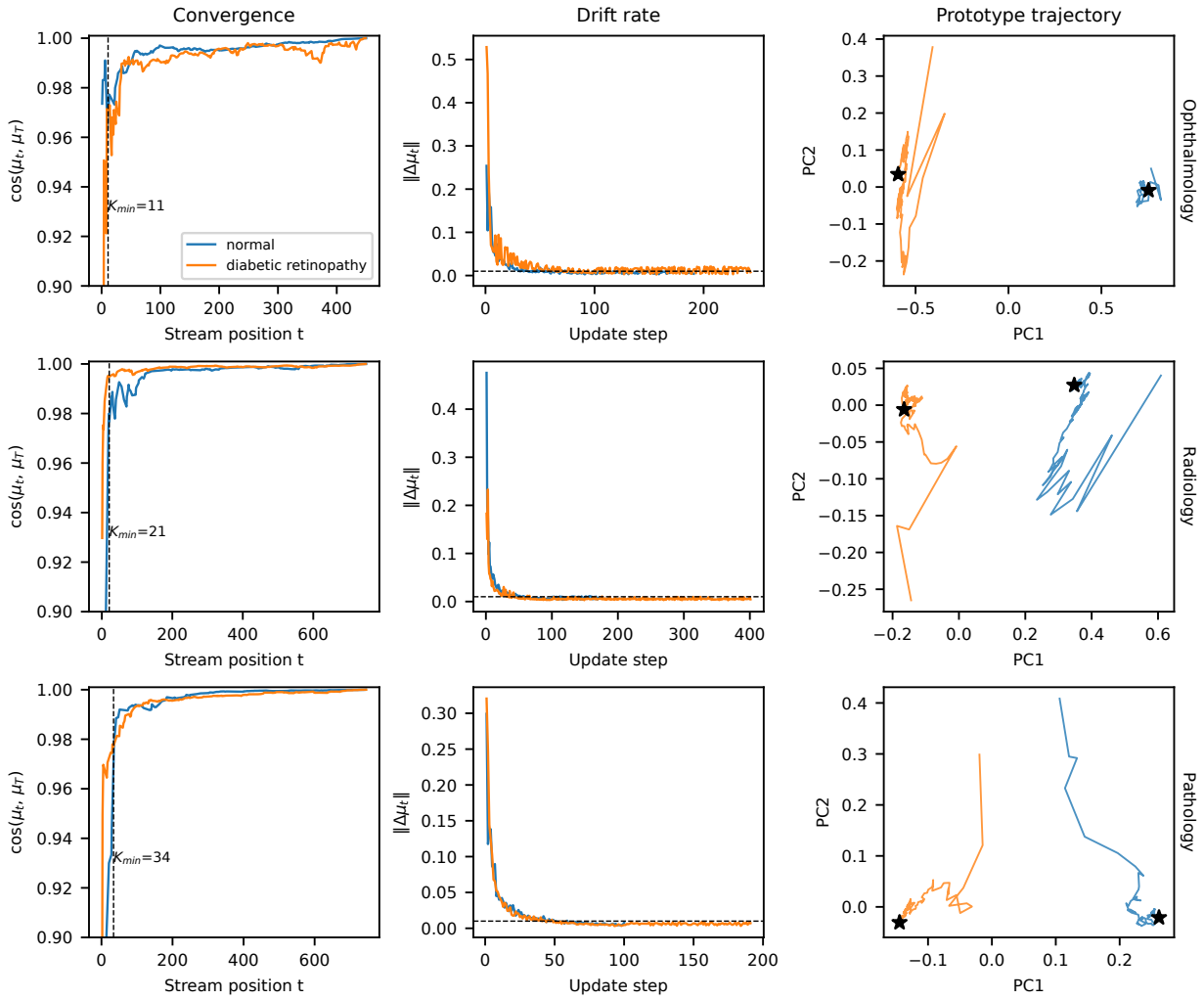



Figure 3: **Prototypes settle within the first hundred admissions.** Cosine similarity of the online prototype to its final value, drift per update, and PCA trajectories on the covariate stream, one row per modality.